Psycholinguistic Diagnostics Revisited with LLMs: A case study on negation

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Abstract

This study builds upon prior research which utilize psycholinguistics stimuli to assess the comprehension of negation in pretrained language models, such as BERT and RoBERTa. We extend the analysis to large language models (LLMs). By examining how LLMs perform under negation, this study aims to provide a broader understanding of their linguistic capabilities and potential limitations. For this purpose, we enrich previous observations on psycholinguistic diagnostics and their extended versions to 80 open-source pretrained language models of different sizes, variations, and generations. We also replicate the previous results with the same models and point out the possible issues in the implementation choices in terms of the evaluation methods. Notwithstanding the vast success of LLMs, our observations show that most LLMs still fail to adjust their behavior under negated contexts in non-conversational examples. The code for the experiments is available on `https://github.com/emrecncelik/llm-negation`.

Introduction

Despite remarkable advances in language modeling, a fundamental linguistic challenge persists: understanding negation. When humans are asked to complete the sentence "A trout is not a __" we can effortlessly understand this negated statement, and complete the sentence with a correct word. However, LLMs often struggle with such seemingly simple linguistic constructions, which can potentially lead to critical errors in real-world applications. In this study, we conduct a comprehensive evaluation, testing 80 diverse language models to determine whether modern LLMs have overcome these fundamental negation challenges.

Limitations of prior work

- Limited model diversity in terms of size, architecture, and training methods (primarily BERT-era models).
- Inconsistent evaluation results due to determiner mismatches and lack of standardized implementation.

Research questions

- How do models of varying sizes, architectures, and training procedures differ in their negation understanding capabilities?
- Does negation insensitivity persist in modern LLMs?

Datasets

All datasets we use come from the work of Ettinger (2020) and prior psycholinguistics experiments. The examples from each datasets are illustrated in Table 1. Since the original datasets are derived from human experiments, their size is notably limited when compared to the large-scale benchmark datasets typically employed for training and evaluating language models. To mitigate this issue, Shivagunde et al. (2023) extended these datasets by artificially generating new examples using GPT-3, and predefined categories and templates as provided in prior psycholinguistics studies (Fischler et al., 1983; Battig and Montague, 1969).

Dataset	Size	Context	$Target_{true}$	$Target_{false}$
NEG-136-SIMP	18 pairs	A trout is a/an A trout is not a/an	fish tool	tool fish
$\text{NEG-1500-SIMP}_{GEN}$	750 pairs	A computer is a/an A computer is not a/an	machine person	person machine
$\text{NEG-1500-SIMP}_{TEMP}$	772 pairs	A cherry is a/an A cherry is not a/an	fruit sport	sport fruit
NEG-136-NAT_{NT}	8 pairs	Old computers may be Old computers may not be	slow fast	fast slow
NEG-136-NAT_{LN}	8 pairs	A baby bunny's fur is very A baby bunny's fur isn't very	soft hard	hard soft

Table 1: Overview of the datasets.

Evaluation

We used the sensitivity definitions from both Ettinger (2020) and Shivagunde et al. (2023), see Figure 1 for more details on how they are calculated. Ettinger's sensitivity is widely known as *direct probability measurement*. The `minicons` library (Misra, 2022) is used to standardize the evaluation. We additionally measured the top-5 accuracy on only the affirmative examples to evaluate the models' hypernym knowledge.

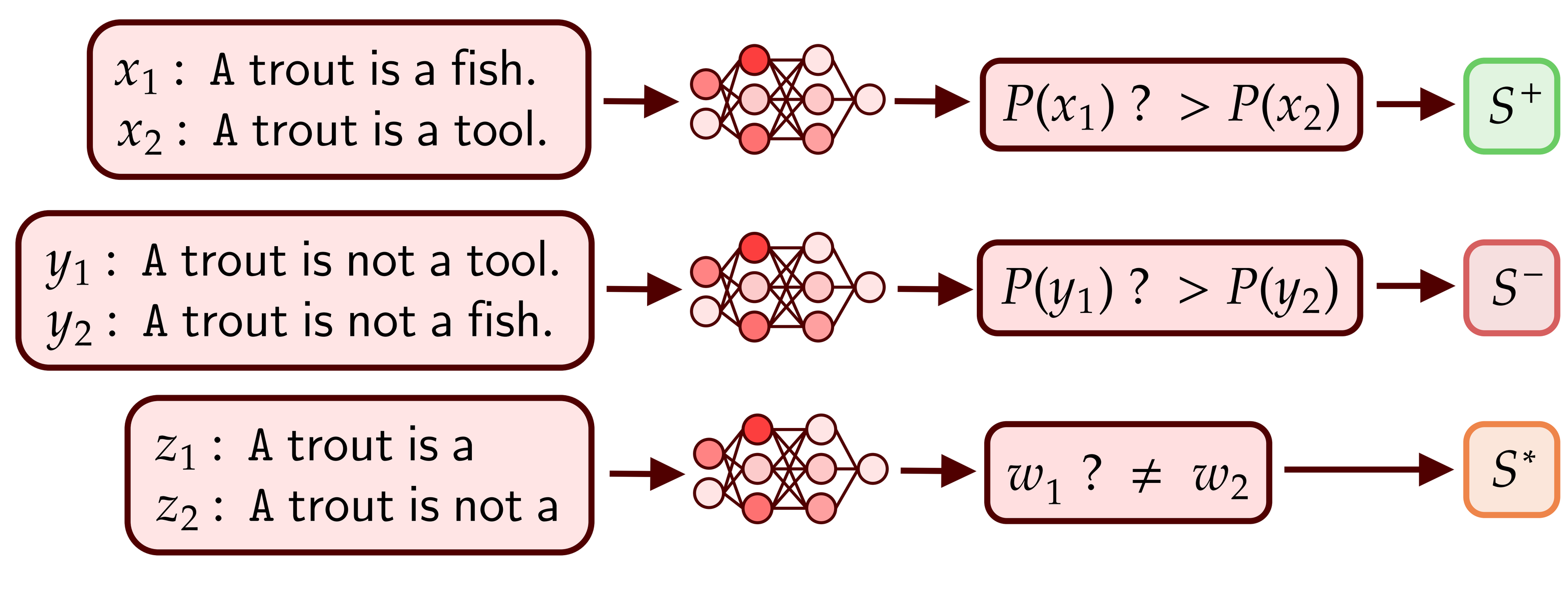

Figure 1: Task formulation and evaluation metrics. S^+ is calculated by comparing the log-probabilities of acceptable affirmative sentence (top). S^- is calculated the same way, but for negated sentences (middle). Finally, S^* is calculated by letting the model finish the incomplete sentence, and seeing if the top-1 prediction has changed or not. (bottom)

Results

	NEG-136-SIMP				$\text{NEG-1500-SIMP}_{TEMP}$				$\text{NEG-1500-SIMP}_{GEN}$			
Model	*Acc.*	$S*$	$S+$	$S-$	*Acc.*	$S*$	$S+$	$S-$	*Acc.*	$S*$	$S+$	$S-$
ModernBERT_{base}	100	38.9	100	0	69.1	44.9	91.2	13.5	60.1	43.2	83.2	27.5
$\text{ModernBERT}_{large}$	100	38.9	100	0	72.7	35.1	90.8	11.7	61.6	47.1	84.9	27.3
NeoBERT	88.9	27.8	100	0	66.0	50.3	92.6	8.6	48.5	46.5	81.7	27.2
EuroBERT_{210M}	83.3	27.8	100	0	60.1	44.3	91.9	10.1	46.9	38.5	80.5	25.5
EuroBERT_{610M}	77.8	22.2	100	0	63.4	45.6	89.6	10.1	39.3	48	79.1	22
$\text{EuroBERT}_{2.1B}$	77.8	33.3	100	5.6	56.9	59.0	86.8	16.2	34.7	48.4	76.5	25.9
Mamba_{130M}	66.7	55.6	94.4	5.6	59.1	48.2	93.5	6.0	42.8	50.5	80.7	20.3
Mamba_{790M}	94.4	44.4	100	0	67.7	47.5	96.1	4.7	53.7	55.2	87.1	21.6
$\text{Mamba}_{1.4B}$	100	38.9	100	0	69.5	41.9	95.6	5.6	54.4	53.7	86.5	21.5
$\text{Mamba}_{2.8B}$	94.4	44.4	100	0	67.5	54.2	95.5	5.3	55.2	65.2	88.8	21.6
Mamba2_{130M}	100	33.3	100	0	65.3	46.9	95.1	5.1	53.2	50.5	86.3	18.8
Mamba2_{780M}	88.9	44.4	100	0	69.2	53.0	95.1	4.4	59.7	59.9	87.6	20.9
$\text{Mamba2}_{1.3B}$	94.4	33.3	100	0	70.1	46.8	95.5	6.1	61.9	61.7	88.9	21.2
$\text{Mamba2}_{2.7B}$	94.4	38.9	100	0	68.7	52.6	95.6	4.0	59.3	69.5	90.7	21.3
RGemma_{2B}	88.9	50	100	0	68.8	50.4	95.7	5.2	60.8	68.8	93.5	19.9
RGemma_{9B}	88.9	55.6	100	0	65.7	58.3	96.1	3.1	61.2	72.9	93.1	21.3
Gemma_{2B}	88.9	50	100	0	65.5	50.9	95.3	5.3	59.7	68.3	92.4	21.1
Gemma_{7B}	83.3	44.4	100	0	64.3	57.0	96.9	4.4	57.9	74	93.1	24.1
Gemma2_{2B}	83.3	55.6	100	0	69.6	49.0	96.5	4.7	64.4	65.6	95.5	19.6
Gemma2_{9B}	83.3	50	100	0	68.7	51.0	96.4	3.8	62.4	67.9	95.7	19.1
Llama2_{7B}	88.9	66.7	100	0	60.8	62.1	97.9	4.2	52.3	78.3	88.1	26.4
Llama3_{8B}	88.9	66.7	100	0	65.7	62.9	95.8	3.6	56.9	79.3	93.6	20.7
Llama3.2_{1B}	94.4	50	100	0	69.4	51.7	94.8	3.0	60.0	68.5	92.3	17.6
Llama3.2_{3B}	94.4	55.6	100	0	66.6	56.4	96.5	3.0	58.1	74.3	92.5	22
Llama3.1_{8B}	88.9	66.7	100	0	65.5	60.1	95.7	3.9	56.0	77.7	93.9	22.4
$\text{Qwen1.5}_{0.5B}$	94.4	50	100	0	65.2	46.5	94.3	5.8	53.6	60.9	84	22.7
$\text{Qwen1.5}_{1.8B}$	94.4	38.9	100	0	68.1	48.8	95.2	4.9	56.0	63.9	88.3	21.9
Qwen1.5_{4B}	94.4	44.4	100	0	66.5	53.1	94.9	6.2	54.5	68.9	88.4	25.2
Qwen1.5_{7B}	88.9	50	100	0	65.3	60.1	95.3	5.8	54	71.5	89.3	26
$\text{Qwen2.5}_{0.5B}$	94.4	38.9	100	0	64.8	47.0	93.2	4.8	50.4	60.7	83.5	21.1
$\text{Qwen2.5}_{1.5B}$	88.9	55.6	100	0	68.1	47.7	96.4	6.2	59.3	67.2	89.7	21.6
Qwen2.5_{3B}	94.4	44.4	100	0	69.0	52.9	95.7	6.0	59.1	71.3	92.3	22.7
Qwen2.5_{7B}	94.4	50	100	0	67.3	58.6	97.0	6.2	56.1	77.3	92.9	25.2
Pythia_{70M}	33.3	55.6	88.9	11.1	23.4	67.5	88.8	13.2	17.9	56.4	76.3	25.1
Pythia_{160M}	50	66.7	100	0	40.3	68.6	90.4	9.4	23.2	68.4	76.4	20.8
Pythia_{410M}	66.7	61.1	100	0	55.5	57.8	91.8	7.3	35.1	57.9	75.6	23.6
Pythia_{1B}	83.3	55.6	100	0	61.8	58.7	93.8	6.9	46.9	64.8	81.9	22.9
$\text{Pythia}_{1.4B}$	94.4	38.9	100	0	64.7	47.4	94.4	4.5	49.7	56.5	84.1	22.5
$\text{Pythia}_{2.8B}$	72.2	50	100	0	65.8	48.7	95.6	5.3	49.9	64.1	84.4	23.1
$\text{Pythia}_{6.9B}$	77.8	44.4	100	0	68.3	55.8	95.7	3.9	52.5	64.4	86.1	23.3
OLMo2_{7B}	94.4	50	100	0	68.1	59.7	97.5	4.2	60.9	74.4	93.6	23.3
RGemma_{2B} IT	88.9	77.8	88.9	16.7	56.9	73.5	85.1	15.8	49.9	79.7	78	22.8
RGemma_{9B} IT	88.9	55.6	94.4	11.1	61.7	64.5	85.6	13.9	55.2	79.3	76.7	20.4
Gemma_{2B} IT	94.4	50	61.1	38.9	64.2	51.4	69.6	39.1	60.5	68	59.6	48.8
Gemma_{7B} IT	61.1	77.8	94.4	5.6	47.8	76.9	80	22.3	34.9	82.1	79.1	23.2
Gemma2_{2B} IT	83.3	61.1	94.4	0	63.8	61.7	83.5	20.1	53.7	73.2	80.7	24.9
Gemma2_{9B} IT	77.8	66.7	88.9	22.2	57.8	72.6	81.9	21.2	49.3	82.9	72.8	31.6
Llama2_{7B} IT	88.9	83.3	100	11.1	55.6	71.4	95.3	7.5	45.6	87.1	81.1	33.6
Llama3_{8B} IT	88.9	77.8	100	5.6	60.3	68.3	95.2	4.8	50.5	82.1	88.4	26.4
Llama3.1_{8B} IT	88.9	77.8	100	0	61.9	64.4	94.9	4.9	53.6	79.3	89.6	25.5
Llama3.2_{1B} IT	94.4	88.9	100	0	64.5	48.2	91.2	7.7	56.5	68.1	77.3	23.1
Llama3.2_{3B} IT	83.3	77.8	100	5.6	59.7	59.4	92.6	7.1	51.3	77.3	83.2	28.7
$\text{Qwen1.5}_{0.5B}$ IT	88.9	66.7	100	0	53.8	61.3	93.0	6.9	43.9	71.9	85.2	16.3
$\text{Qwen1.5}_{1.8B}$ IT	88.9	50	100	0	66.1	57.5	94.0	3.9	50.7	70.8	83.3	21.5
Qwen1.5_{4B} IT	72.2	66.7	100	0	50.4	51.9	93.5	7.1	35.7	58.0	84.7	17.2
Qwen1.5_{7B} IT	83.3	77.8	100	5.6	61.6	65.3	94.3	8.1	50.9	78.8	89.6	30.5
$\text{Qwen2.5}_{0.5B}$ IT	94.4	44.4	100	0	65.5	51.0	93.5	4.9	50.4	64.3	81.6	21.9
$\text{Qwen2.5}_{1.5B}$ IT	88.9	50	100	0	67.7	45.1	96.4	6.1	57.7	63.1	88.8	23.1
Qwen2.5_{3B} IT	94.4	44.4	100	0	68.6	56.1	94.9	6.9	57.7	71.2	90.4	24.4
Qwen2.5_{7B} IT	88.9	55.6	100	0	65.8	63.2	96.2	7.3	55.2	80	91.6	29.1
OLMo2_{7B} SFT	88.9	50	100	0	67.5	58.3	94.4	6.2	53.7	71.7	80.7	27.2
OLMo2_{7B} DPO	77.8	72.2	94.4	0	64.4	64.7	93.0	6.5	50.7	75.7	80.5	28.8
OLMo2_{7B} IT	77.8	66.7	94.4	0	64.0	64.4	93.1	6.9	50.8	76.9	80.1	28.5

Table 2: Comprehensive experiment results on NEG-SIMP datasets.

1. All tested models regardless of size, training methods, or architectures fail to assign greater probability to the correct completion in negated sentences from NEG-SIMP and variations.
2. Model size has no effect on S^+ and S^-, while it improves S^*. Meaning larger models are better able to change their top-1 prediction, however the false completions still rank very high.
3. Prior claims that larger datasets showing a much better negation sensitivity incorrectly attributed metric differences to dataset size; our analysis shows very minor improvement regardless of dataset scale.
4. Increase in S^-, and decrease in S^+ in extended datasets are overlapping, suggesting this change is due to poorer knowledge of the extended hypernyms rather than negation sensitivity. This is supported by drastic the decrease in accuracy.
5. Models show sensitivity to negation in natural contexts, parallel to the findings of Ettinger.

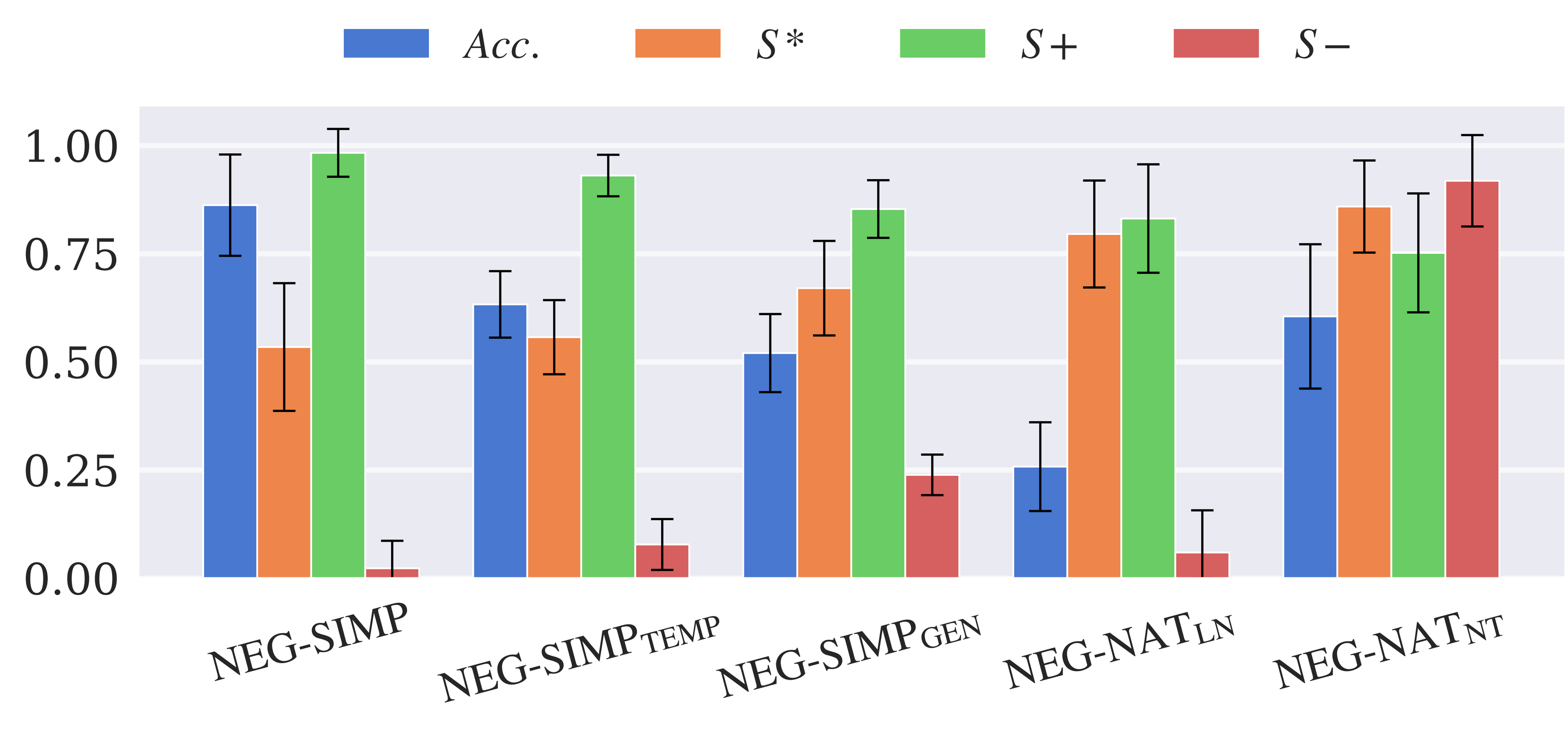

Figure 2: Summary of the metrics from all experiments. The bars and black lines indicate the mean and standard deviation respectively.

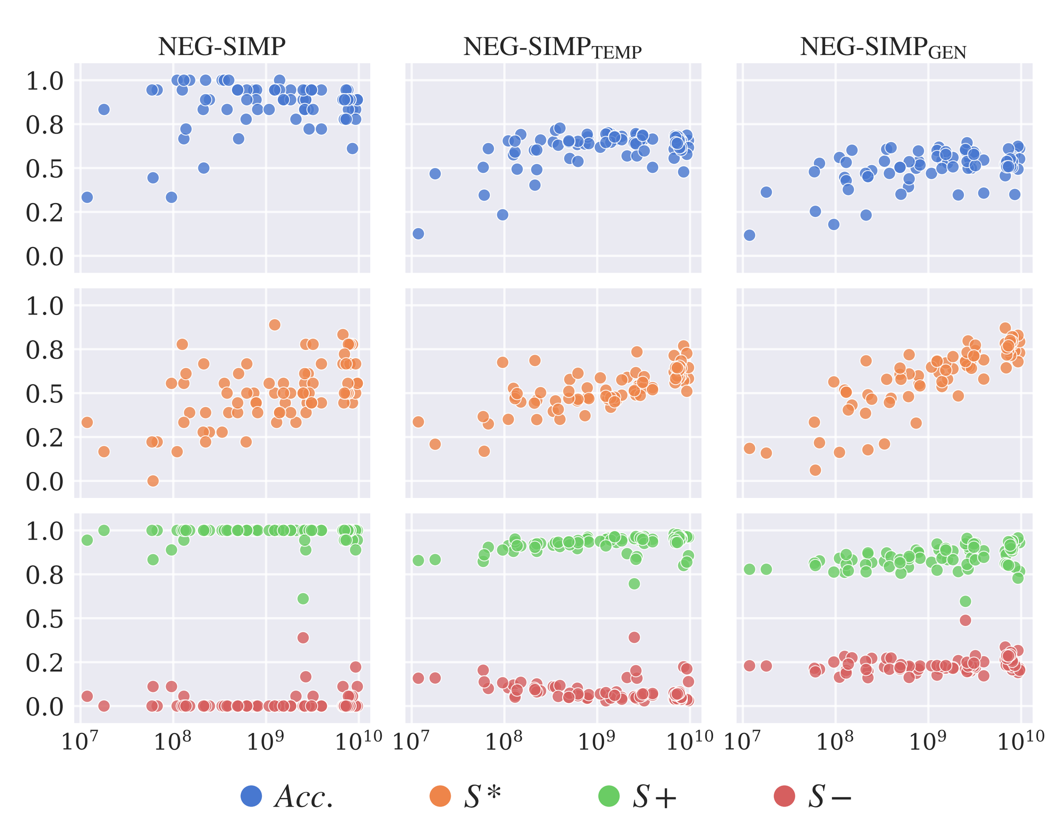

Figure 3: Performance metric by size figure for all 80 models on NEG-SIMP dataset and its variations. The model size is shown in logarithmic scale.

Scan for references.